

Aqua Probe

The Aqua Security Runtime Input Validator Tool

User Guide



Introduction



Aqua Probe is an interactive command-line tool for validating Aqua Security runtime controls in Kubernetes environments.

It guides users through repeatable tests, deploys a known test container, and points security teams to the Aqua Console audit workflow for review.

What Is Aqua Probe?

- A bash-based runtime security input validator for Linux and macOS operators connected to a Kubernetes cluster.
- Checks Kubernetes connectivity, Aqua Enforcer daemonsets, and the Aqua Kube-Enforcer deployment.
- Deploys the Aqua Probe test container image: **ericgomes56/aqua-probe:1.0**.
- Provides guided prompts for runtime, assurance, privilege, volume, port, DNS/IP reputation, and resource policy tests.

Supported Features

Aqua Probe menu options

- Deploy/Redeploy test container
- Real-time Malware Protection [Delete action]
- Drift Prevention
- Block Cryptocurrency Mining
- Block Fileless Execution
- Reverse Shell
- Executables Blocked
- Block Container Exec
- Bad DNS/IP Reputation
- File Block
- Package Block
- Port Scanning Detection
- Block Non-compliant Images
- Block Unregistered Images
- File Integrity Monitoring
- System Integrity Monitoring
- Limit Container Privileges
- Block Non-Kubernetes Containers
- Port Block
- Volumes Blocked
- Block Non-compliant Resources

Usage

- Ensure **kubectI** is configured for the target Kubernetes cluster.
- Run **.aqua-probe.sh** from a terminal.
- Select option 1 to deploy or redeploy the test container.
- Choose a security control option and follow the prompt.
- Review results in the Aqua Console under **Security Reports** -> **Audit**.

```
Explore various security features of Aqua Security with this interactive tool.
Experience Real-time Malware Protection, Drift Prevention, and much more.
Get ready to dive into container security with Aqua!

=====

Checking Kubernetes connection.....[✓] Done
✓ Kubernetes cluster connected

Checking Aqua Enforcer (aqua-agent) daemonset.....[✓] Done
✓ Aqua Enforcer daemonset found

[!] Aqua test container does not exist - Select option 1 to deploy.

Proceeding to menu...

Please select an option:
1. Deploy/Redeploy test container
2. Test Real-time Malware Protection [Delete action]
3. Test Drift Prevention
4. Test Block Cryptocurrency Mining
5. Test Block Fileless Execution
6. Test Reverse Shell
7. Test Executables Blocked [ps]
8. Test Block Container Exec
9. Terminate Program

Enter your choice (1-9): 3

[!] In order to test out the use case successfully, please ensure that the following prerequisites are met:
1. Create a Custom Policy with Drift Prevention Control enabled
2. Ensure that the Custom Policy is set to 'Enforce' mode
3. Ensure that Block Container Exec Control is disabled

Proceed? (y/n):
```

Requirements

- Bash shell on Linux or macOS.
- **kubectl** configured to connect to a Kubernetes cluster.
- Aqua Enforcer daemonset deployed in the Kubernetes cluster.
- Aqua Kube-Enforcer deployment named **aqua-kube-enforcer** for admission-control validation.
- Internet access to a Docker registry, or the ability to push **ericgomes56/aqua-probe:1.0** to a local registry.
- Permissions to deploy pods, deployments, and policy-validation resources in the Kubernetes cluster.

Running Aqua Probe

Standard execution:

```
./aqua-probe.sh
```

```
./aqua-probe.sh --no-instructions
```

Aqua Security

```
=====
Welcome to Aqua Probe, the Aqua Security Runtime Input Validator Tool
=====
```

```
Checking Kubernetes connection.....[✓] Done
✓ Kubernetes cluster connected
```

```
Checking Aqua Enforcer daemonset.....[✓] Done
✓ Aqua Enforcer daemonset found
```

```
Checking Aqua Kube-Enforcer deployment.....[✓] Done
✓ Aqua Kube-Enforcer deployment found
```

Please select an option:

1. Deploy/Redeploy test container
2. Test Real-time Malware Protection [Delete action]
3. Test Drift Prevention
4. Test Block Cryptocurrency Mining
5. Test Block Fileless Execution
6. Test Reverse Shell
7. Test Executables Blocked
8. Test Block Container Exec
9. Test Bad DNS/IP Reputation
10. Test File Block
11. Test Package Block
12. Test Port Scanning Detection

```
./aqua-probe.sh --no-instructions
```

```
./aqua-probe.sh --no-instructions
```

```
./aqua-probe.sh --image <image_name>
```

```
./aqua-probe.sh --image <image_name>
```

```
./aqua-probe.sh --namespace <value>
```

```
./aqua-probe.sh --namespace <value>
```

```
./aqua-probe.sh --daemonset <value>
```

```
./aqua-probe.sh --daemonset <value>
```

```
./aqua-probe.sh --help      ./aqua-probe.sh --version
```

```
./aqua-probe.sh --help      ./aqua-probe.sh --version
```



Step-by-step Guide

- Run **`./aqua-probe.sh`**.
- Aqua Probe checks Kubernetes connectivity.
- Aqua Probe checks the Aqua Enforcer daemonset.
- Aqua Probe checks the Aqua Kube-Enforcer deployment named **`aqua-kube-enforcer`**.

Step-by-step Guide

- Select option 1 to deploy or redeploy the Aqua Probe test container.
- Use **ericgomes56/aqua-probe:1.0** by default, or pass **--image** for a local registry image.
- Select one of the security control test options from the menu.

```
> ./aqua-probe.sh --image ericgomes56/aqua-probe:1.0 --no-instructions

Tool developed by Eric Gomes, Senior DevSecOps Architect @ Aqua Security

=====
Welcome to Aqua Probe, the Aqua Security Runtime Input Validator Tool
=====

Checking Kubernetes connection.....[✓] Done
✓ Kubernetes cluster connected

Checking Aqua Enforcer daemonset.....[✓] Done
✓ Aqua Enforcer daemonset found

Checking Aqua Kube-Enforcer deployment.....[✓] Done
✓ Aqua Kube-Enforcer deployment found

Please select an option:
1. Deploy/Redeploy test container
2. Test Real-time Malware Protection [Delete action]
3. Test Drift Prevention
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```

Prerequisite Prompts

- Each feature explains the Aqua policy or control expected for that test.
- Some assurance and admission-control tests require policy settings before deployment.
- Use **--no-instructions** only when the environment has already been prepared.

Example prompts include Assurance Policy checks for malware, unregistered images, non-compliant resources, and runtime controls such as drift, file block, package block, port block, volumes blocked, and DNS/IP reputation.

Execute A Test

- Aqua Probe runs the selected command or Kubernetes deployment automatically after confirmation.
- Runtime examples include **curl https://68.183.212.246**, **cat /etc/passwd**, **tar**, port-scanning attempts, and file integrity operations.
- Kubernetes validation examples include privileged containers, host namespaces, blocked volumes, ports, and non-compliant resources.

Review Results

- After a test is blocked or audited, open the Aqua Console.
- Go to **Security Reports -> Audit**.
- Review the security incident generated by the selected Aqua Probe control test.
- Repeat with another menu option to validate additional controls, or select option 22 to terminate the program.

Download Aqua Probe

- GitHub Repository: <https://github.com/ericgomes56/aqua-probe>
- Test Container Image: docker.io/ericgomes56/aqua-probe:1.0
- Script: `./aqua-probe.sh`





Thanks

Aqua Probe